The documentation and process conversion measures necessary to comply with this amendment shall be completed by 26 May 1998

INCH-POUND

MIL-PRF-19500/181F
AMENDMENT 1
26 February 1998

PERFORMANCE SPECIFICATION

SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON, LOW-POWER
TYPES 2N718A, 2N1613, AND 2N1613L; JAN, JANTX, AND JANTXV

This specification is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 5

4.3, screen table, screen 11, delete "I_{CBO1}" and substitute "I_{CBO2}".

4.3, screen table, screen 13, delete "ΔI_{CBO1}" and substitute "ΔI_{CBO2}".

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Table I, subgroup 2, delete the first two tests:

Breakdown voltage collector-to-base	3001	Bias condition D, I_C = 100 µA dc, pulsed (see 4.5.1)	$V_{(BR)CBO}$	75		V dc	
Breakdown voltage emitter-to-base	3026	Bias condition D, I_E = 100 µA dc, pulsed (see 4.5.1)	$V_{(BR)EBO}$	7		V dc	

and substitute the following tests:

Collector-to-base cutoff current	3036	Bias condition D, V_{CB} = 75 V dc, pulsed (see 4.5.1)	I_{CBO1}		10	µA dc	
Emitter-to-base cutoff current	3061	Bias condition D, V_{EB} = 7 V dc, pulsed (see 4.5.1)	I_{EBO1}		10	µA dc	

Table I, subgroup 2, collector to base cutoff current, symbol column; delete : "I_{CBO1}" and substitute "I_{CBO2}".

Table I, subgroup 2, Emitter-base cutoff current, symbol column; delete : "I_{EBO}" and substitute "I_{EBO2}".

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Table I, subgroup 3, collector to base cutoff curent, symbol column; delete : "I_{CBO2}" and substitute "I_{CBO3}".

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Table II, steps 1 and 2, symbol column; delete : "I_{CBO1}" (two places) and substitute "I_{CBO2}".(two places).

The attached insertable replacement pages listed below are replacements for stipulated pages. When the new pages have been entered in the document, insert the amendment as the cover sheet to the specification.

Replacement page	Page replaced
3	3
4	4

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Custodians:
Army - CR
Navy - EC
Air Force - 17
NASA - NA

Review Activities:
Army - AR, AV, MI, SM
Navy - AS, CG, MC
Air Force - 13, 19, 85, 99

Preparing Activities:
DLA - CC

(Project 5961-1962)

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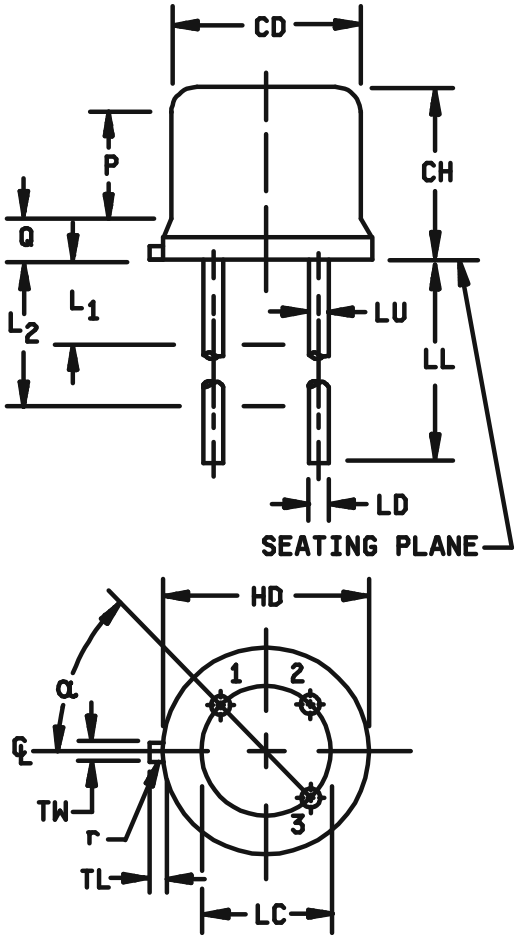
Symbol	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
CD	.178	.195	4.52	4.95	
CH	.170	.210	4.32	5.33	
HD	.209	.230	5.31	5.84	
LC	.707 Nom		1.796 Nom		5
LD		.021		0.53	8
LL	.500	.750	12.7	19.05	
LU	.016	.019	0.41	0.48	4, 8
L_1		.050		1.27	
L_2	.250		6.35		
TL	.028	.048	0.71	1.22	5
TW	.036	.046	0.91	1.17	
P	.100		2.54		3, 6
Q		.040		1.02	
r		.007		0.18	
α	45° TP		45° TP		



NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. This zone is controlled for automatic handling. The variation in actual diameter within this zone shall not exceed .010 inch (0.254 mm).
4. (Three leads) LU applies between L_1 and L_2. LD applies between L_2 and .5 inch (12.70 mm) from seating plane. Diameter is uncontrolled in L_1 and beyond .5 inch (12.70 mm) from seating plane.
5. Measured from maximum diameter of the actual device.
6. Details of outline in this zone optional.
7. The collector shall be electrically connected to the case.
8. In accordance with ANSI Y14.5M, diameters are equivalent to ∅x symbology.
9. Lead number 1 - emitter; lead number 2 - base; lead number 3 - collector.

FIGURE 1. Physical dimensions (TO-18).

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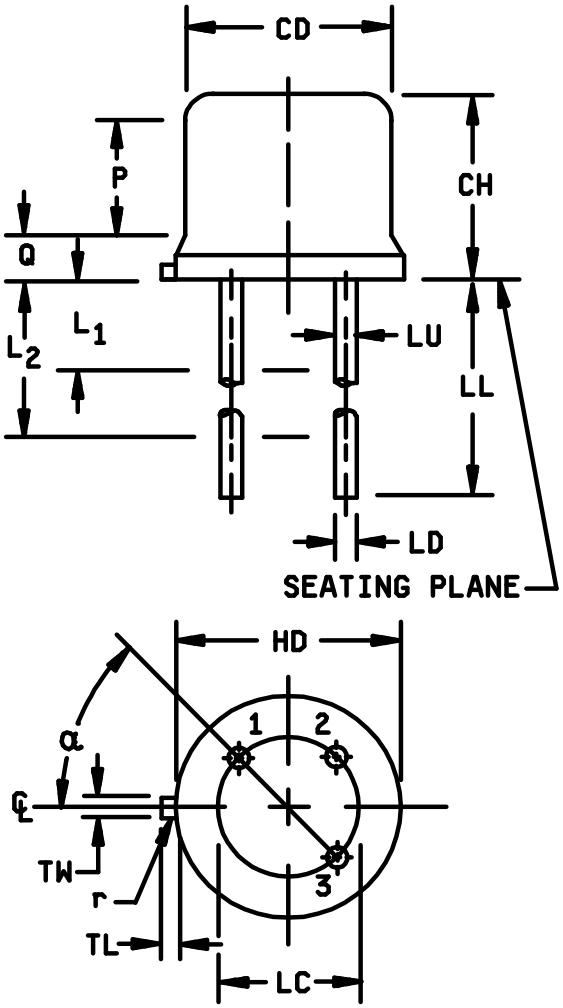
Symbol	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
CD	.305	.335	7.75	8.51	
CH	.240	.260	6.10	6.60	
HD	.335	.370	8.51	9.40	
h	.009	.041	0.23	1.04	
LC	.200 TP		5.08 TP		7
LD	.016	.021	0.41	0.53	8, 9
LL	See notes 12 and 13				
LU	.016	.019	0.41	0.48	8. 9
L_1		.050		1.27	8, 9
L_2	.250		6.35		8, 9
P	.100		2.54		6
Q		.050		1.27	5
r		.010		0.25	
TL	.029	.045	0.74	1.14	4
TW	.028	.034	0.71	0.86	3
α	45° TP		45° TP		7



NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Beyond r (radius) maximum, TW shall be held for a minimum length of 0.011 inch (0.28 mm).
4. TL measured from maximum HD.
5. Outline in this zone is not controlled.
6. CD shall not vary more than .010 inch (0.25 mm) in zone P. This zone is controlled for automatic handling.
7. Leads at gauge plane .054 +.001, -.000 inch (1.37 +0.03, -0.000 mm) below seating plane shall be within .007 inch (0.18 mm) radius of true position (TP) at a maximum material condition (MMC) relative to the tab at MMC. The device may be measured by direct methods or by the gauge and gauging procedure described on gauge drawing on figure 2.
8. LU applies between L_1 and L_2. LU applies between L_2 and LL minimum. Diameter is uncontrolled in L_1 and beyond LL minimum.
9. All three leads.
10. The collector shall be electrically and mechanically connected to the case.
11. r (radius) applies to both inside corners of tab.
12. For transistor types 2N1613, dimension LL is .500 inch (12.70 mm) minimum, and .750 inch (19.05 mm) maximum.
13. For transistor types 2N1613L, dimension LL is 1.500 inches (38.10 mm) minimum, and 1.750 inches (44.45 mm) maximum.
14. Lead number 1 - emitter; lead number 2 - base; lead number 3 - collector.

FIGURE 2. Physical dimensions (TO -5 and TO -39).

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